Note 14.8 Lagrange Multipliers

Introduction

Lagrange multipliers is the method for finding extreme values of constrained functions.

Constrained Maxima and Minima

Substitution

Most simple method but not always useful to choose appropriate variables to substitute.

The Method of Lagrange Multipliers

For the function $f(x, y, z)$ whose variables are subject to a constraint $g(x, y, z) = 0$, to find the local maximum and local minimum values,find the values of x, y, z, λ on the surface $g = 0$ among the points where

$$\nabla f = \lambda \nabla g$$

and λ is Lagrange multiplier.

The two equations $g = 0$ and $\nabla f = \lambda \nabla g$ are enough to solve all unknowns.(consider possibly different cases)

We can understand $f(x, y, z) = 0$ and $g(x, y, z) = 0$ as the level curve. That is, f changes until it is tangent to g.

Lagrange Multipliers with Two Constraints

To find the local extrema of the function f constrained by two functions $g_1(x, y, z) = 0, g_2(x, y, z) = 0$, we need to find the x, y, z, λ, μ such that

$$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2, g_1(x, y, z) = 0, g_2(x, y, z) = 0$$

We also understand the three function as level curves. And the constraint functions intersect on a curve and we need to find the extrema on the intersection curve. Then we need to change f until the curve is the tangent to f. That is, ∇f is normal to the curve. Since $\nabla g_1, \nabla g_2$ are also normal to the curve, $\nabla f, \nabla g_1, \nabla g_2$ are on the same plane.

Exercises

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